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REPUBLIQUE DU CAMEROUN



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INTERNSHIP REPORT

DESIGN and IMPLEMENTATION OF DAILY MONEY COLLECTION SYSTEM CASE MONITRACK

Internship done from 24th of June to 30th of August 2024 in partial fulfillment of the HND
(Higher National Diploma)

Field of study: **Computer Engineering**

Specialty: **software engineering**

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ACADEMIC YEAR

2024-2025

CERTIFICATION

This is to certify that an internship with the theme: “ **DESIGN AND IMPLEMENTATION OF A DAILY MONEY COLLECTION WEB APPLICATION** ”: case of Monitrack , is originally written and presented by **MAGNE NDE STEPHANIE SHELIA** after a two month of internship carried out at HONOWA TECH to meet up with the partial requirement and regulation governing the award of the Higher National Diploma (HND) of Institut Universitaire de la Cote in the field software engineering.

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DEDICATION

TO MY FAMILY

ACKNOWLEDGEMENTS

The accomplishment of this work required the assistance of some people. Failing to quote them individually here, they receive my most sincere thanks. We are practically grateful to :

- Mr. GUIMEZAP Paul, Founder and President of IUC for the establishment of this structure for the purpose of training student.
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- Mr. FOMAZOU TCHINDA, my academic supervisor, who have been a source of inspiration and motivation for me. Always providing me with task that helped me through my journey.
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- ALL MY TEACHERS for my knowledge they installed in me.
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- To everyone I have failed to mention here, thank you.

ABSTRACT

Daily money collection is an activity carried by microfinance it is an integral part of human history and modern society. It is a system that enables individuals and businesses to manage their finances and engage in economic activities. The technological advancements of the 19th century has made this to become easier to handle, with the aim of laying and emphasis on the reliability ,security and a smooth management of the banking system which is made up of creating bank accounts,saving money withdrawing money seeking for loans and loan reimbursement for the client and creating clients accounts managing clients transactions as well as validation of loans depending on the clients credibility, we decided to built a web application that will handle these transaction in a user friendly manner and without major effort and taking into consideration security constraints.

RESUME

La collecte journalière d'argent est une activité menée par la microfinance, elle constitue une part intégrante de l'histoire humaine et de la société moderne. Il s'agit d'un système qui permet aux individus et aux entreprises de gérer leurs finances et de participer à des activités économiques. Les avancées technologiques du XIX^e siècle ont facilité la gestion de ces activités, avec pour objectif de mettre l'accent sur la fiabilité, la sécurité et une gestion fluide du système bancaire, qui inclut la création de comptes bancaires, l'épargne, le retrait d'argent, la recherche de prêts, le remboursement des prêts pour les clients, ainsi que la création de comptes clients, la gestion des transactions des clients et la validation des prêts en fonction de la crédibilité des clients. Nous avons décidé de développer une application web qui gèrera ces transactions de manière conviviale, sans effort majeur et en tenant compte des contraintes de sécurité.

PREFACE

With the aim of ensuring a suitable development and providing business with competent and competitive workforce in various fields, the Cameroon government through the ministry of higher education has opened the private institution of education. This then created the HIGER INSTITUTE OF TECHNOLOGY AND INDUSTRIAL DESIGN (ISTDI) by decree N° 02/0094/MINESUP/DDES/ESUP of September 13 2002 and authorization of opening N°01/02/MINESUP/DDES/ESUP of 18 September 2002, located in the coastal region, Wouri department, district of Douala 5th, in Logbessou district. The ISTDI then created the UNIVERSITY INSTITUTE OF THE COAST by decree N° 5/05156/N/MINESDUP/DDES/ESUP/SAC/EBM. University Institute of the Coast (IUC) consist of five (5) different institution which are:

❖ School of Engineering and Applied Science

It consists of:

- HND INDUSTRIAL PROGRAMS
 - Electrical power systems (EPS)
 - Industrial Computing and Automation (ICA)
 - Telecommunication (TEL)
 - Software Engineering (SWE)
 - Network Security (NWS)
 - Chemical Manufacturing (CM)
 - Mechanical Manufacturing (MEM)
 - Building Science and Technology (BST)

- HND COMMERCIALS PROGRAMS
 - Marketing (MKT)
 - Management (MGT)

- Accountancy (ACC)
- Banking and Finance (BF)
- Logistic and Transport Management (LTM)
- BTS PROGRAMS
 - GENIE CIVILE (GC)
 - GEOMETRE TOPOGRAPH (GT)
 - BATIMENTS
 - TRAVEAUX PUBLIC (TPU)
- BACHELORS OF ENGINEERING (B-Eng)
 - Computer Science Engineering
 - Electromechanical Engineering
 - Civil and Environmental Engineering
- BACHELORS OF TECHNOLOGY (B-Tech)
 - Chemical Engineering
 - Electrical and Electronics Engineering
 - Civil Engineering
 - Software Engineering
 - Mechatronics Engineering
 - Industrial Automation and Computing
 - Welding and Fabrication Technology
 - Thermal Engineering Technology
- BACHELORS OF HEALTH SCIENCE (B-HES)
 - Physiotherapy
 - Medical Laboratory
 - Nursing Science
- LIPRO
 - Genie Civil
- MPRO

➤ Genie Civil

2. L'institut Supérieur des Technologies et du Design Industriel (ISTDI)

It consist of :

❖ **BTS PROGRAMS**

- Génie Chimique des procédés (GCP)
- Maintenance des systèmes informatique (MSI)
- Informatique Industriel (II)
- Electrotechnique (ET)
- Electronique (EN)
- Froid et climatisation (FC)
- Maintenance et après-vente automobile (MAVA)
- Bâtiment (BAT)
- Travaux publics (TP)
- Géomètre topographe (GT)
- Energie Renouvelable (ER)
- Génie Bois (GB)
- Chaudronnerie (CH)
- Fabrication Mécanique (FM)

❖ **DSEP PROGRAM :**

- Réseaux et Télécommunication (RT)

❖ **LISCENCES PROFESSIONNELLES INDUSTRIELLES** in partnership with the university of DSCHANG: VIII

- Administration des systèmes Réseaux
- Génie Logiciel
- Automatique et Informatique Industriel
- Electrotechnique
- Electronique
- Management des systèmes Automobiles
- Maintenance et Expertise des Automobiles
- Maintenance des systèmes Industriels

- Génie Énergétique et Industriel
- Production et commercialisation du bois
- Génie Mécanique et productique

❖ MASTER PROFESSIONNEL COMMERCIAL in partnership with the University of DSCHANG

- Génie Electrique et informatique Industriel
- Génie des Réseaux et Télécommunications
- Systèmes d'information et de Génie Logiciel
- Systèmes d'information Audit et Conseil

3. L'institut de Commerce et d'Ingénierie d'affaire (ICIA)

❖ BTS PROGRAMS

- Action Commercial
- Assurance
- Banque et Finance
- Communication d'Entreprise
- Commerce International
- Gestion Logistique et Transport
- Informatique de Gestion

❖ LISCENCES PROFESSIONNELLES COMMERCIALES ET GESTION in partnership with the University of DSCHANG:

- Assurance
- Banque
- Banque – Assurance
- Comptabilité Control et Audit
- Finance – Comptabilité
- Gestion des Ressources Humaines
- Logistique et Transport
- Marketing
- Management des Opération du Commerce International IX

❖ MASTER PROFESSIONNEL COMMERCIAL in partnership with the University of DSCHANG:

- Finance – Comptabilité
- Audit Comptable et Financier
- Banque
- Fiscalité
- Management des Organisation
- Marketing – Communication
- Gestion des Ressources Humaines
- Logistique et Stratégique

❖ MASTER ISUGA – France in partnership with EMBA France

- EXECUTIVE MBA in partnership with ESSEC Douala

4. L'institut d'Ingénierie Informatique d'Afrique Centrale (3IAC)

❖ Cycle des TIC in partnership with CCNB-DIEPPE of CANADA

- Réseautique et sécurité
- Programmes et Analyse

❖ MASTER EUROPEAN in partnership with 3IL France

- Genie Logiciel
- Administration des Systèmes Réseaux

❖ MASTER PROFESSIONNEL in partnership with ENSP Yaoundé

- Génie Civil
- Génie Énergétique et Environnement
- Génie Industriel et Maintenance

❖ CYCLE INGENIEUR Information (3IL-France)

- Pôle de Recherche Innovation et Entrepreneuriat

5. LES PROGRAMMES INTERNATIONAUX DES SCIENCE ET TECHNOLOGIES DE L'INNOVATION (PISTI)

- ❖ Cycle des Classes Préparatoire au Grandes Ecoles d'Ingenieurs(CP) in partnership with the University of MAINE France
 - Classes préparatoires
 - Licence Sciences et technique
- ❖ CYCLE INGENIEUR de Génie Industriel (Polytechnique- NANCY France)
- ❖ INGENIEUR BIOMEDICAL in partnership with the University of Tor Vergata (ROME II)
- ❖ ARCHITECTURE ET DESIGN INDUSTRIEL in partnership with the University of CAMERINO (ROME I)

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LIST OF ACRONYMS

WRITTEN AND PRESENTED BY:MAGNE NDE STEPHANIE SHELIA

HND: Higher National Diploma

IUC : Institut Universitaire de la Cote

HONOWA-TECH : Honowa Technologies

CSS: Cascading Style Sheet

DBMS: Database Management System

OOP: Object Oriented Programming

PHP: Hypertext Pre-processor

SDLC: System Development Life Cycle

SWE: Software Engineering

SQL: Structured Query Language

UML: Unified Modelling Language

XAMPP: Cross-platform, Apache, MySQL, PHP, and Perl

UX/UI: User Experience /User Interface.

GENERAL INTRODUCTION

The technological advancements of the 21st century have significantly transformed various sectors of our daily lives. Computers and digital technologies have become essential tools for accomplishing a wide array of tasks. In light of this progress, we have developed a web application specifically designed to manage money transactions within a banking framework, facilitating the creation and management of accounts. Given the sensitivity of financial transactions often referred to as the "root of all conflict" it is crucial to handle these processes with care and precision. Our web application aims to streamline the management of financial transactions, ensuring secure and efficient operations. By providing a reliable platform for daily money collection, we aspire to contribute to the overall enhancement of public service delivery in Cameroon, addressing existing challenges and improving user experience in financial management.

To achieve our goal, we will present on one hand the internship which is divided into part as seen below:

PART 1: CONCEPTUAL FRAME WORK

Chapter 1: presentation of the company activities carried out during the internship period.

Chapter 2: activities carried out during the internship period.

PART 2: EMPIRICAL STUDY

Chapter 3: diagnostic of existing system of Monitrack.

Chapter 4: Implementations of proposed system.

GENERAL CONCLUSION

COST AND ESTIMATION

REFERENCES

PART ONE : CONCEPTUAL FRAME WORK

CHAPTER ONE: PRESENTATION OF HONOWA AND INTERNSHIP ACTIVITIES

1.1 PRESENTATION OF HONOWA TECHNOLOGY

In this section, we will explore Honowa Technologies, a pioneering force in the field of innovative solutions. We will delve into the company's mission, key offerings, and the impact of its technologies on various sectors. Additionally, we will examine the underlying principles that drive Honowa Technology's commitment to excellence and innovation. Through this overview, we aim to provide a comprehensive understanding of how Honowa Technology is shaping the future of technology and its applications.

1.1.2 MISSION OF HONOWA-TECH

HONOWA-Tech is a technological enterprise involved in . its main mission is to improve the performance and profitability of businesses and organizations through tailored digital and technological solutions.

1.1.1 HISTORY OF HONOWA TECHNOLOGY

HONOWA Technologies is a company founded by DOUNLA TABOUTSA CARLOS in 2019. To this day, its founder remains its Managing Director. The company is a business software publisher for various fields of activity and has integrated digital marketing into its services.

Our passionate, young, and experienced team is dedicated to exceeding client expectations, approaching each project with enthusiasm and expertise.

1.1.3 SERVICES OFFERED BY HONOWA

HONOWA-Tech offers the following services;

- application development(mobile,desktop and web).
- Digital marketing
- Training center for both activities.

1.1.4 LOCATION OF HONOWA-TECH

It is located in the Littoral Region of Cameroon, in the town of Douala specifically at Bepanda, near the Seventh Police Station.

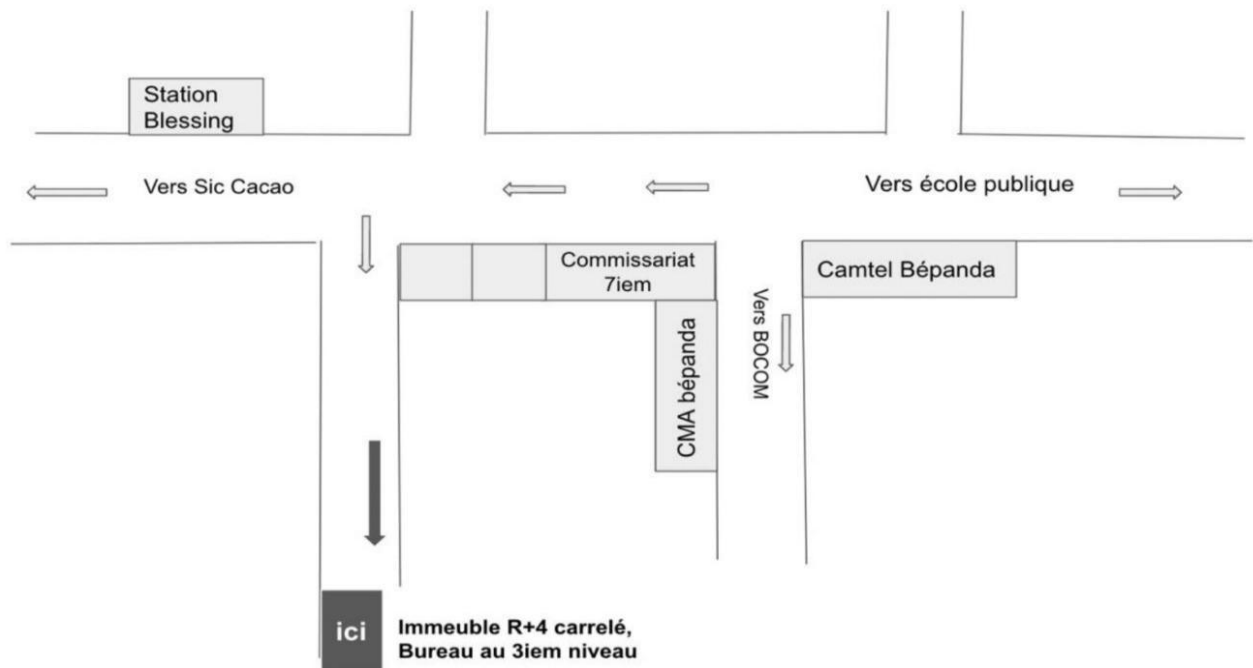


Figure 1: Location of HONOWA-TECH

1.1.5 ORGANISATIONAL STRUCTURE OF HONOWA-TECH

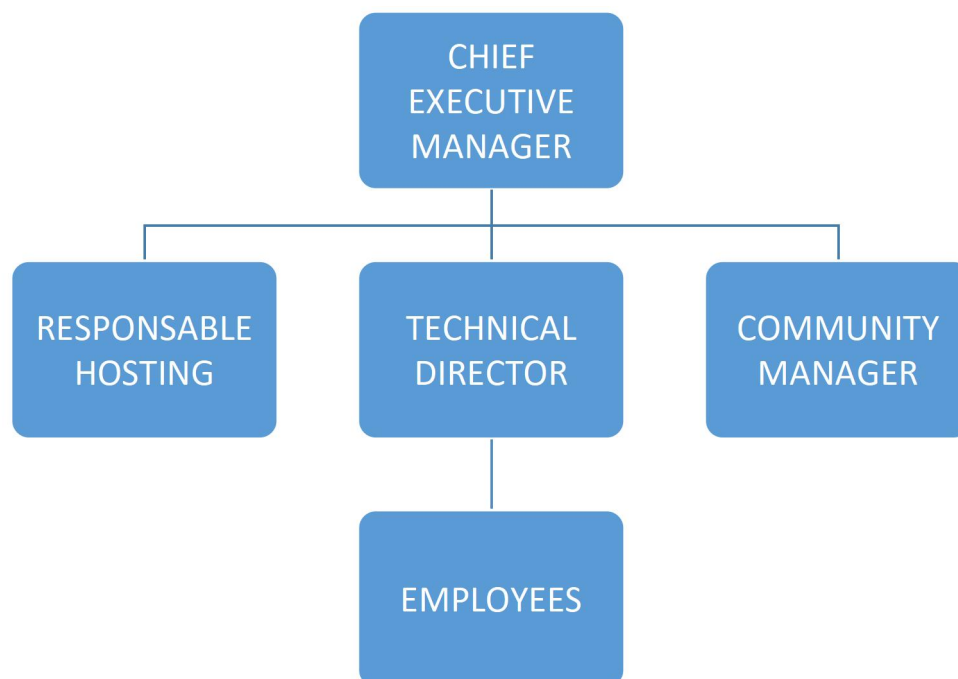


Figure2:Organizational Chart of HONOWA-TECH

1.1.6 DETAIL ON VARIOUS DEPARTMENTS

❖ **Honowa-business center:**

- Digital marketing.

- Development of web applications.
- Development of mobile applications.

❖ **Honowa-Training center:**

- Programming and web development.
- Computer networking (Cisco).
- Digital marketing.

1.2 INTERNSHIP ACTIVITIES

This section outlines the various activities I undertook during my internship at Honowa-Tech. The experience provided me with valuable insights into the practical applications of my academic knowledge, as well as opportunities to develop essential skills in a professional environment. From learning programming languages to engaging in web application development, each activity contributed to my growth and understanding of the technology sector. Below, I will detail the specific tasks I completed, the challenges I faced, and the skills I acquired throughout my internship.

1.2.1 Introduction

Upon my arrival at Honowa-Tech, I was warmly welcomed by Mr. Dounla Carlos, the CEO and my professional supervisor. He provided a thorough briefing on the company's mission, activities, and the expectations for my internship experience. Following this introduction, I was given a set of guidelines to follow, and my work commenced immediately.

PERIOD	PERFORMED ACTIVITIES	DIFFICULTIES ENCOUNTERED	SOLUTIONS PROPOSED	SKILL ACQUIRED
from June 24 to June 29	-Introduction to company and other interns in my field	-No major problem was encountered	-No problem hence no solution required	-Socialization skills -Punctuality -Team work -Respect of hierarchy
From July 1 to July 13	Learning programming languages: HTML, CSS, JAVASCRIPT	-No problem faced	- No problem encountered hence no solution	-New programming skills in

From july 19 To july 25	Designing the homepage of Honowa Technologies	-No problem faced	- No problem encountered hence no solution	Ability to apply theoretical skills
From July 31 to August 10	Revision of the MVC and design of a university information system	-No problem faced	- No problem encountered hence no solution	Understanding basic concepts of data modeling
From august 5 to august 23	Learning PHP and Laravel Framework	-No problem faced	- No problem encountered hence no solution	Ability to manage a database
From august 12 to august 23	Developing a library management application	Debugging difficulties	Performing unit and integration testing	Ability to debug codes
From august 25 to august 30	Choosing the theme and drafting the specifications document	-No problem faced	- No problem encountered hence no solution	Learning to draft a specifications document

Table 1: Activities carried out during internship

1.2.2 JUSTIFICATION OF TOPIC

During my internship at Honowa-Tech, we focused on developing websites and web applications. As a result, the company proposed the creation of a web application specifically designed for daily money collection. This solution aims to address the shortcomings of existing systems, ultimately enhancing their effectiveness and operational efficiency. By participating in this project, I was able to contribute to a meaningful initiative that has the potential to significantly improve financial processes within the industry.

PART TWO : EMPERICAL STUDY

CHAPTER TWO: DIAGNOSTIC OF EXISTING SYSTEM

2.1: INTRODUCTION

A daily money collection management system that automates the operations of a daily money collect institutions, improving efficiency, accuracy and customer satisfaction.

Scope: The system will handle customer management, transaction management.

2.2 PRESENTATION OF THE SYSTEM USED BY Monitrack

Monitrack is a daily money collect enterprise that offers money collection services hence welcomes a good number of clients everyday and workers in monitrack.

Workers also called agents are the one's going to meet clients to registered transactions (either deposit or withdrawal) in their register (a book) where behave to put the client information and the amount of money he is giving.

2.2.1 DATA FLOW DIAGRAMS

Data Flow Diagrams (DFDs) are a visual representation of the flow of data within a system. They are used primarily in the fields of software engineering and systems analysis to illustrate how information moves between processes, data stores, and external entities. DFDs help stakeholders understand the relationships and interactions between different components of a system, making them valuable tools for both system design and analysis.

A DFD typically consists of four main components: processes, data stores, data flows, and external entities. Processes represent the operations or transformations that data undergoes within the system, while data stores indicate where data is stored for future use. Data flows are arrows that depict the movement of data between processes, data stores, and external entities, which are sources or destinations of data outside the system. This clear visual structure allows for easy identification of how data is processed and where it is stored.

Here are some data flow diagram representing the flow of information in Monitrack

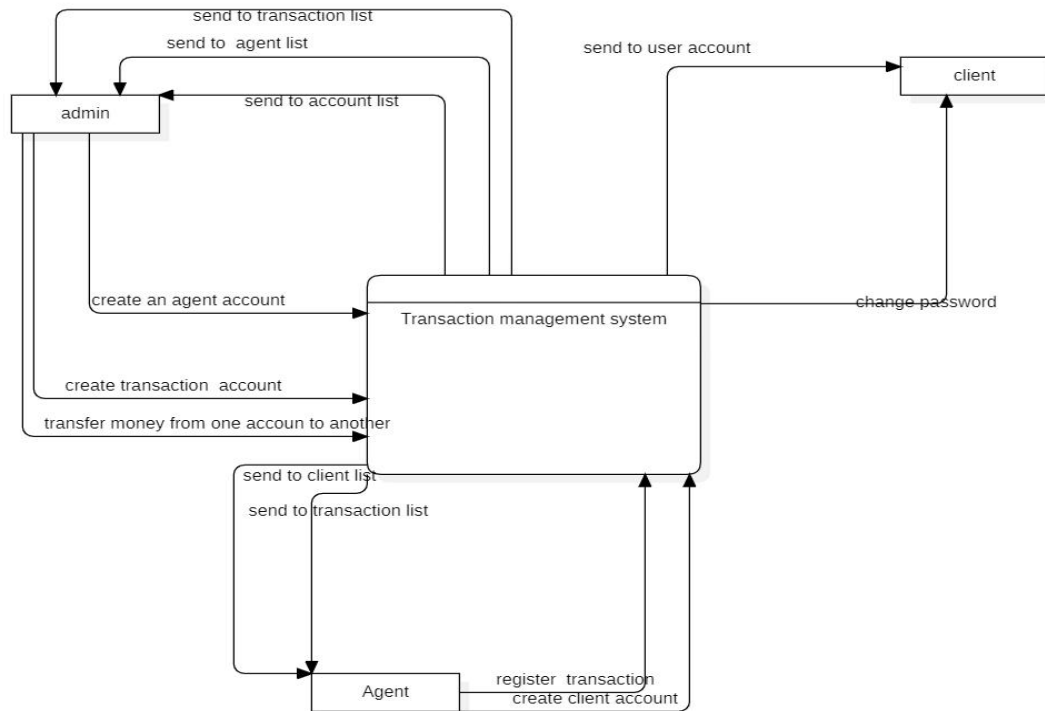


Figure 3: Level zero data flow diagram

- The **Admin** can create agent accounts and initiate transactions.
- **Agents** are responsible for registering transactions and creating client accounts.
- **Clients** can change their passwords and send requests to view their transaction lists.
- The system maintains a **Transaction List** for all operations performed.
- Money transfers can occur between different accounts, managed through the system.
- The interactions ensure a smooth flow of information between users and the transaction management system

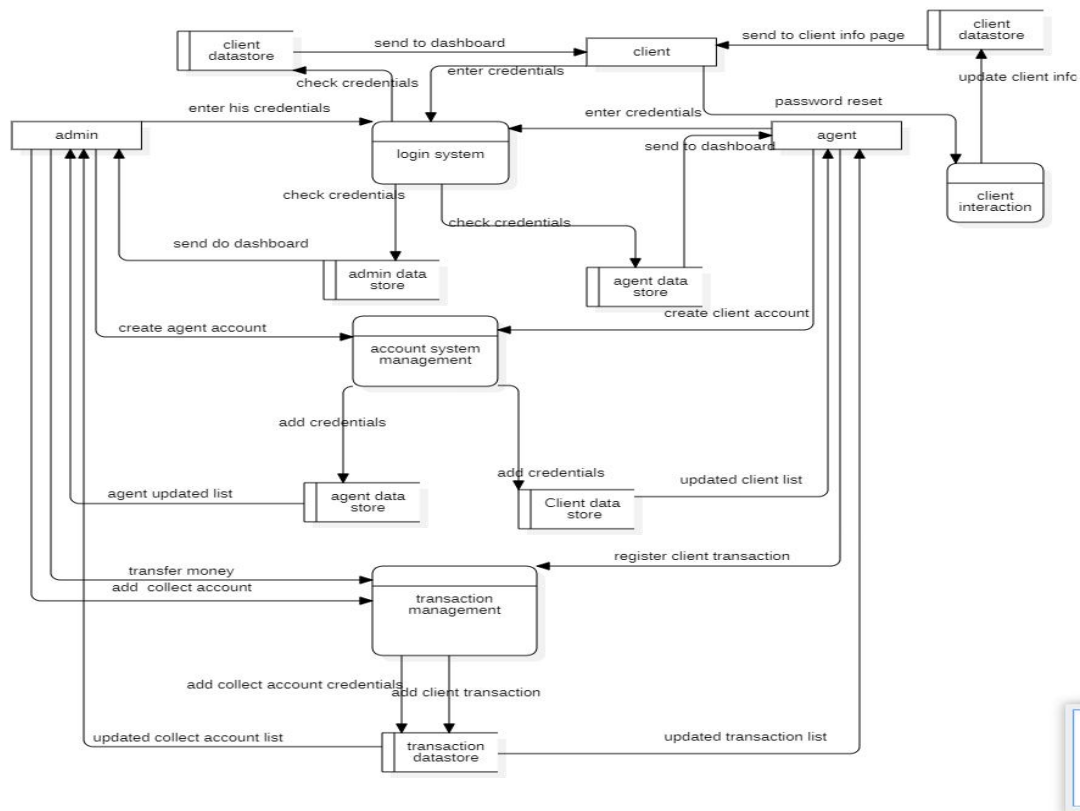


Figure 4: Level one data flow diagram

- The **Admin** manages user accounts, including creating agent accounts and updating credentials.
- **Clients** log in using their credentials to access the system dashboard.
- The **Agent** interacts with the client through the dashboard, facilitating account creation and updates.
- The system maintains a **Data Store** for agent and client credentials.
- Transactions are managed through a **Transaction Management** process, allowing money transfers between accounts.
- All changes are reflected in the respective data stores and dashboards, ensuring data consistency.

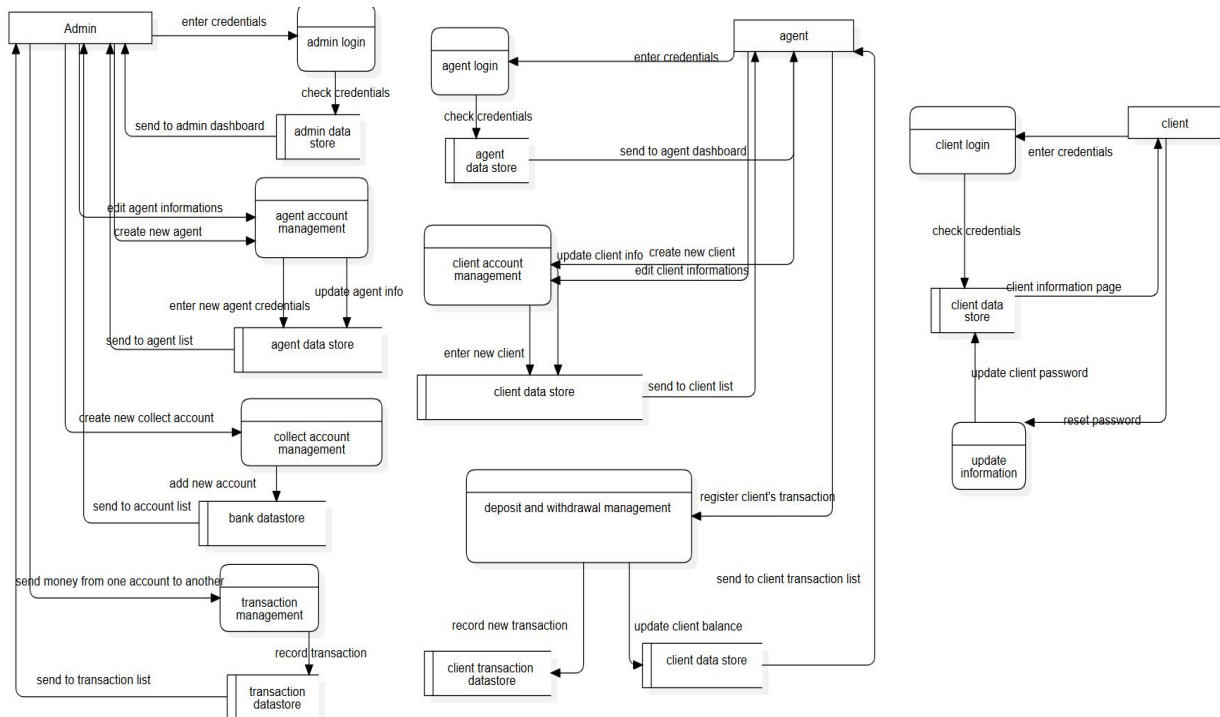


Figure 5: Level two data flow diagram



Figure 6: Existing system of monitrack

2.3 RESEARCH METHOD

In the course of my research, I used several methods to identify the problems:

- **INTERVIEW:** I asked the administrator of Monitrack and he gave me information of the about thier system and show me the register it workers are using to register trasanctions.
- **OBSERVATION:** During the free period I had while being at agent working place , I could see how they are interacting with clients.
- **INTERNET:** The internet provided me with essential information about what I needed and great solutions and manner of approach in order to solve the vulnerabilities of the existing system monitrack .

2.4 ADVANTAGES OF THE PRESENT SYSTEM

- Lower Initial Costs:
 - Manual systems do not require significant investment in software, hardware, or IT infrastructure.
 - This can be advantageous for small banks or startups with limited budget
- No Dependency on Technology:
 - Manual systems are not affected by technical issues like software bugs, system crashes, or power outages.
 - This can be beneficial in areas with unreliable internet or electricity.

2.5 LIMITATIONS OF THE PRESENT SYTEM

- Higher risk of Human Errors: Manual data entry and calculations are prone to mistakes,such as incorrect figures,duplicate entries or omitted information.
- Time consuming :manual system require significant time for data-entry.calculations,records and record keeping. Transactions like deposits and withdrawals and loan processing take longer to complete,leading to delays in service
- lack of real time data: manual system cannot provide real-time updates on account balances ,transactions or other critical information. This lead to delays in decision-making and customer dissatisfaction

2.6 CHARACTERISTICS OF THE NEW SYSTEM

The limitations of existing systems, as seen in institutions like **Monitrack** , highlight the need for a new, improved daily money collection System. The proposed system will:

- **Fully Automate Processes:** Reduce errors and save time by automating loan applications, repayments, and reporting.
- **Provide Real-Time Updates:** Enable stakeholders to access real-time information for better decision-making.
- **Offer a User-Friendly Interface:** Improve usability for customers, loan officers, and admins.
- **Ensure Scalability:** Handle large volumes of data and growing user bases.
- **Enhance Security:** Protect sensitive data with robust encryption and authentication mechanisms.

CHAPTER THREE:ANALYSIS AND DESIGN OF MONITRACK

In order to bring forward an effective solution, a thorough analysis had to be made. The analysis phase is the first critical phase in software development. Therefore, in this chapter, I will present a complete analysis of the system I have proposed, stating the methodology I will use, requirements, design and implementation of the system.

3.1 GENERAL PRESENTATION OF THE METHOD OF ANALYSIS

3.1.1 SYSTEM DEVELOPMENT LIFE CYCLE

SDLC is a term used in Information systems and Software Engineering which refers to the process for building an application. It is also called Software Development Process. It is subdivided into the Planning, Analysis, Design, Coding, Testing and Development, depending on the methodology used.

Some methodology are :waterfall,iterative,V-shape,agile and prototype.

3.2 PLANNING PHASE

This phase is also known as the requirement gathering phase and is one of the most important and fundamental stages in the SDLC of a project. It consists of drafting a work plan and carrying out detail feasibilities and preliminary studies about what is to be done and involves studying the present system if any is in place and need for amelioration. For the case of my project, the pen and paper system was used to manage activities. There was no computerized system that could facilitated work-from-home task management, hence I decided to build a web application that will facilitate Monitrack in their activities (daily money collection). I gathered my requirements using non-structured interview, observation and internet research.

3.2.2 REQUIREMENTS GATHERING

3.2.2 .1 Actors of the system

Actors of the system :these are the people that would use the system

- Admin:CEO of monitrack
- Agent:workers at monitrack

- Clients of monitrack

3.2.2.2 FUNCTIONAL REQUIREMENT

Functional requirements of monitrack are those activities that the system can do. As seen on the table below. Functional requirements describe the specific features and functionalities of your system. These are derived from the needs of the stakeholders identified in your analysis.

Identifier	User story
REQ-1	AS an admin, I can create an agent account
REQ-2	As an admin, I can create and associate accounts
REQ-3	As an admin, I can transfer money from one account to another
REQ-4	AS an agent or admin, I can create client account
REQ-5	As an agent or admin, I can register client transaction
REQ-6	As a client, I can see my account balance and transaction
REQ-7	As a user, I change my information

After requirement analysis

Identifier	User story
REQ-1a	As an admin ,I can delete agent account
REQ-1b	As an admin, I can update agent information's
REQ-4a	As an admin, I can delete client account
REQ-4b	As an admin or an agent, I can update client informations

Table 3: USER STORY TABLE

3.2.2.3 NON-FUNCTIONAL REQUIREMENT

The non-functional requirements are those requirements that describes how the system should perform rather than what they should do.

- **Performance:** The system should handle up to 1,000 concurrent users.
- **Security:** Data encryption and secure user authentication.
- **Usability:** User-friendly interface for both staff and customers(clients).

- **Scalability:** The system should be scalable to accommodate future growth.

Below is the summarized non functional requirements of the system

Category	Requirements
Performance	The system should handle up to 1,000 concurrent users.
Scalability	The system should scale to support up to 10,000 users.
Security	All sensitive data must be encrypted using AES-256 encryption.
Availability	The system should have an uptime of 99.9%.
Usability	The system should have an intuitive user interface.
Maintainability	The system should support for easy updates.
Backup	The system should perform daily automated backups.

Table 4: NON FUNCTIONAL REQUIREMENT TABLE

3.3 SYSTEM ANALYSIS

This phase serves to set the stage and bound problems. It involves breaking down the system into different pieces to analyse the situation, analysing the project goals, breaking down what needs to be created and seeing how the system can be created without affecting the company negatively and provide more efficient solution to problems faced.

3.3.1 JUSTIFICATION ON METHODOLOGY USED

For this project I used the Agile Methodology. The agile model is an SDLC model used for Software development. Agile is a software development lifecycle (SDLC) model that emphasizes flexibility, collaboration, and iterative progress. It is designed to accommodate changing requirements and improve efficiency by delivering functional software in small, incremental pieces, known as iterations or sprints. It is made up of seven phases.

1. Requirement Gathering and Analysis:

Here, requirements are gathered from users using various techniques, such as interviews, surveys, and workshops. Analysts evaluate whether the requirements are feasible and if the project goals can be achieved within budget constraints.

2. Design Phase:

This phase involves establishing the technical requirements (e.g., programming languages, data structures, services) needed to fulfill the requirements identified in the previous phase. The design outputs include logical and physical diagrams, and the design may represent a new system or modifications of an existing one.

3. **Implementation:**

During implementation, the designs are translated into code using appropriate programming languages. This phase transforms the design specifications into functional software. Once developed, the software is installed on users' machines for testing and use.

4. **Testing:**

Throughout the development process, testing is conducted to ensure the software meets quality standards. This can include unit testing, integration testing, and user acceptance testing, often occurring concurrently with implementation.

5. **Deployment:**

This phase involves releasing the completed software to users. Deployment may include training users and providing documentation to support the transition.

6. **Maintenance:**

After deployment, the software enters the maintenance phase, where bugs and issues are identified and resolved. This phase may also involve updating the software to adapt to new requirements or improve performance.

7. **Review or Evaluation:**

After deployment, teams conduct reviews or evaluations to gather feedback from users and stakeholders. This feedback is essential for continuous improvement and informs future iterations.



Figure 7: Agile methodology

3.3.2 MODELLING LANGUAGE

The modelling language I used to implement this application is the **Unified Modelling Language**. it is a general purpose developmental modelling language in the software engineering field which intends to provide a standard way to visualize the design of a system.

3.3.3 MODELLING TOOLS

For this system, we used;

- Star UML: We built our UML diagrams(use case,data flow diagrams,class diagrams) using star UML.
- FIGMA:we designed the graphical user interface using Figma
- Edrawmax :We built our UML diagrams(sequence diagrams ,state chart diagram) using Edrawmax.
- CANVA :for the application logo .

3.4 DESIGN PHASE

3.4.1 USE CASE

A use case is a list of actions or events typically defining the interactions between a role and a system to achieve a goal. The role is known in **UML** as an actor and may be human or an external system.

A use case diagram consists of the **boundary** which defines the system of interest in relation to the world around it, the actors which are usually individuals involved in the functioning of the system, the use cases which are specific roles played by the actors within and around the system and finally the **relationships** between and among the actors and the use cases.

3.4.1.1 USE CASE ANALYSIS

❖ ACTORS GOAL LIST

Initiator	Initiator's goal	participants	Use case name
Admin	Create an agent account	Database	Add Agent
	Modify agent information	database	Edit Agent info
	Delete agent	Database	DeleteAgent
	Create new coffer account to store money	database	Create Bank Account
	Transfer money from one coffer account to another account	Database	transfermoney
	Delete client account	database	Delete Client
Agent and admin	Create client account	database	Add client
	Register client transactions	Database,agent,client	Transaction
	Modify client information	Database	Edit Client
Client	See his/her transaction and account balance	Database	Transactionlist
	Change his/her password	database	Password reset

Table 5: ACTOR GOAL LIST TABLE

❖ USE CASE DESCRIPTION

Use case name	Add agent
Actors	Admin
Actor's goal	Register a new agent
Preconditions	➤ The admin is on the daily money collect management system.
Flow of Events for Main Success Scenario	1. Login 2. Click on Agent management and select New agent. 3. Admin fill the form with the required informations and click on save.
Flow of Events for Main Alter Scenario	The form is send back to the admin in case of invalid credentials Form rejected if The agent with same credentials exist

	already.
Postconditions	➤ The new agent information is successfully stored in the database. ➤ Admin is sent to agent list
Assumptions	The actor is in front of the interface.

Table 6: ADD AGENT USE CASE

Use case name	Edit agent info
Actors	Admin
Actor's goal	Edit an agent information
Preconditions	➤ The admin has valid login credentials. ➤ The system is operational, and the database is accessible
Flow of Events for Main Success Scenario	1. Admin has to login 2. Click on agent management and choose the option List. 3. The admin select the agent he want to edit information and click on edit. 4. A form with pre-filled credentials on the selected Agent is presented to the admin. 5. After modification clicks on save.
Flow of Events for Main Alternative Scenario	➤ Form rejected due to invalid credentials.
Postconditions	➤ Agent information is successfully stored in the database ➤ Admin is send to Agent list
Assumption	The admin is in front of the interface.

TABLE 7: EDIT AGENT INFORMATION USE CASE

Use case name	Delete agent account
Actors	Admin
Actor's goal	Delete an agent account

Preconditions	➤ The admin has valid login credentials.
Flow of Events for Main Success Scenario	1. Admin has to login 2. Click on agent management and choose the option List. 3. The admin select the agent he want to delete. 4. The admin recieve a pop up message for the confirmation of the delete and click on yes.
Flow of Events for Alternative Scenario	NO ALTERNATIVE SCENARIO
Postconditions	➤ The Agent status is change to inactive in the database. ➤ Admin is sent to Agent list .
Assumption	The admin is infront of the interface

Table 8: DELETE AGENT USE CASE

Use case name	Bank
Actors	Admin
Actor's goal	It consist of creating an account to store money
Preconditions	➤ The admin has to be on the daily money collect management system
Flow of Events for Main Success Scenario	1. Admin has to login. 2. Choose finance and on the list choose Account then new. 3. Admin have to fill the form with the valid credentials . 4. Submit the form by clicking on save.
Flow of Events for Alternative Scenario	➤ Form rejected incase of invalind credentials credentials. ➤ Form rejected incase account with same credentials exist already in the system.
Postconditions	➤ Coffor account is successfully stored in the database. ➤ Admin is sent to coffer account list.
Assumption	The admin is infront of the interface.

TABLE 9: CREATE BANK ACCOUNT USE CASE

Use case name	UC-5:transfer Money
Actors	Admin
Actor's goal	Transfer money from one coffer account to another
Preconditions	➤ The admin has valid login credentials. ➤ The account's(source and destination) must exist
Flow of Events for Main Success Scenario	1. Admin has to login. 2. Choose finance and on the list choose Transfer then new. 3. Admin have to fill the form with the valid credentials . 4. Submit the form by clicking on save.
Flow of Events for Alternative Scenario	Form rejected due to invalid invalid credentials
Post-conditions	➤ The transfer credentials are successfully stored in the database ➤ Admin is redirect to the transfer list
Assumption	The admin is infront of the interface.

TABLE 10: TRANSFER MONEY USE CASE

Use case name	UC-6:delete client
Actors	Admin,Agent
Actor's goal	Delete a client account
Preconditions	➤ The admin or Agent has valid login credentials.
Flow of Events for Main Success Scenario	5. Admin or agent has to login 6. Click on Client management and choose the option List. 7. The admin select the client he want to delete. 8. The admin recieve a pop up message for the confirmation of the delete and click on yes.

Flow of Events for Alternative Scenario	NO ALTERNATIVE SCENARIO
Postconditions	➤ The client status is change to inactive in the database. ➤ Admin or agent is sent to clientlist .
Assumption	The admin or agent is infront of the interface

TABLE 11: DELETE CLIENT ACCOUNT USE CASE

Use case name	UC-7:Add client
Actors	Admin,Agent
Actor's goal	Register a new client
Preconditions	➤ The admin or Agent is on the daily money collect management system.
Flow of Events for Main Success Scenario	1. Login 2. Click on Agent management and select New agent. 3. Admin fill the form with the required information and click on save.
Flow of Events for Main Alter Scenario	The form is send back to the admin or agent in case of invalid credentials Form rejected if The client with same credentials exist already.
Postconditions	➤ The new client information is successfully stored in the database. ➤ Admin or egnt is sent to client list
Assumptions	The admin or agent is infront of the interface.

TABLE 12: ADD CLIENT USE CASE

NAME	UC-8:Transaction
ACTORS	Admin or Agent

Actor's goal	Register a client transaction
Preconditions	➤ The admin or agent has valid login credentials. ➤ The system is operational, and the database is accessible ➤ Client must have enough money in hi account in case it is the transaction is a withdrawal.
Flow of Events for Main Success Scenario	1. Login 2. Click on Transaction and select New . 3. Admin fill the form with the required information and click on save.
Flow of Events for Main Alter Scenario	The form is send back to the admin or agent in case of invalid credentials
Postconditions	➤ The Transaction information is successfully stored in the database. ➤ Client balance and money in the coffer account are updated ➤ Admin or agent is sent to transaction list
Assumptions	The admin or agent is infront of the interface.

TABLE 13: TRANSACTION USE CASE

Use case name	UC-9:Edit Client information
Actors	Admin,Agent
Actor's goal	Edit a client information
Preconditions	➤ The admin or agent has valid login credentials. ➤ The system is operational, and the database is accessible
Flow of Events for Main Success Scenario	6. Admin or agent has to login 7. Click on agent management and choose the option List. 8. The admin or agent select the client he want to edit information and click on edit. 9. A form with pre-filled credentials on the selected client is

	presented to the admin or agent. 10. After modification clicks on save.
Flow of Events for Main Alternative Scenario	➤ Form rejected due to invalid credentials.
Postconditions	➤ client information is successfully stored in the database ➤ Admin or agent is send to client list
Assumption	The admin or agent is in front of the interface.

TABLE 14: EDIT CLIENT INFORMATION USE CASE

NAME	UC-11:Reset password
ACTORS	Admin,client and agent
Actor's goal	Change his password
Preconditions	The actor is on the daily money collect management system.
Flow of Events for Main Success Scenario	1. Admin or agent has to login 2. Click on profile and choose edit password. 3. Actor fill the form with the new password and and confirm password. 4. After modification clicks on save.
Flow of Events for Alternative Scenario	Form rejected in case new password and confirm password are not same
Post-conditions	➤ Password is successfully updated in the database ➤ User is send to the dashbord
Assumption	The actor is in front of the interface.

TABLE 15:RESET PASSWORD USE CASE

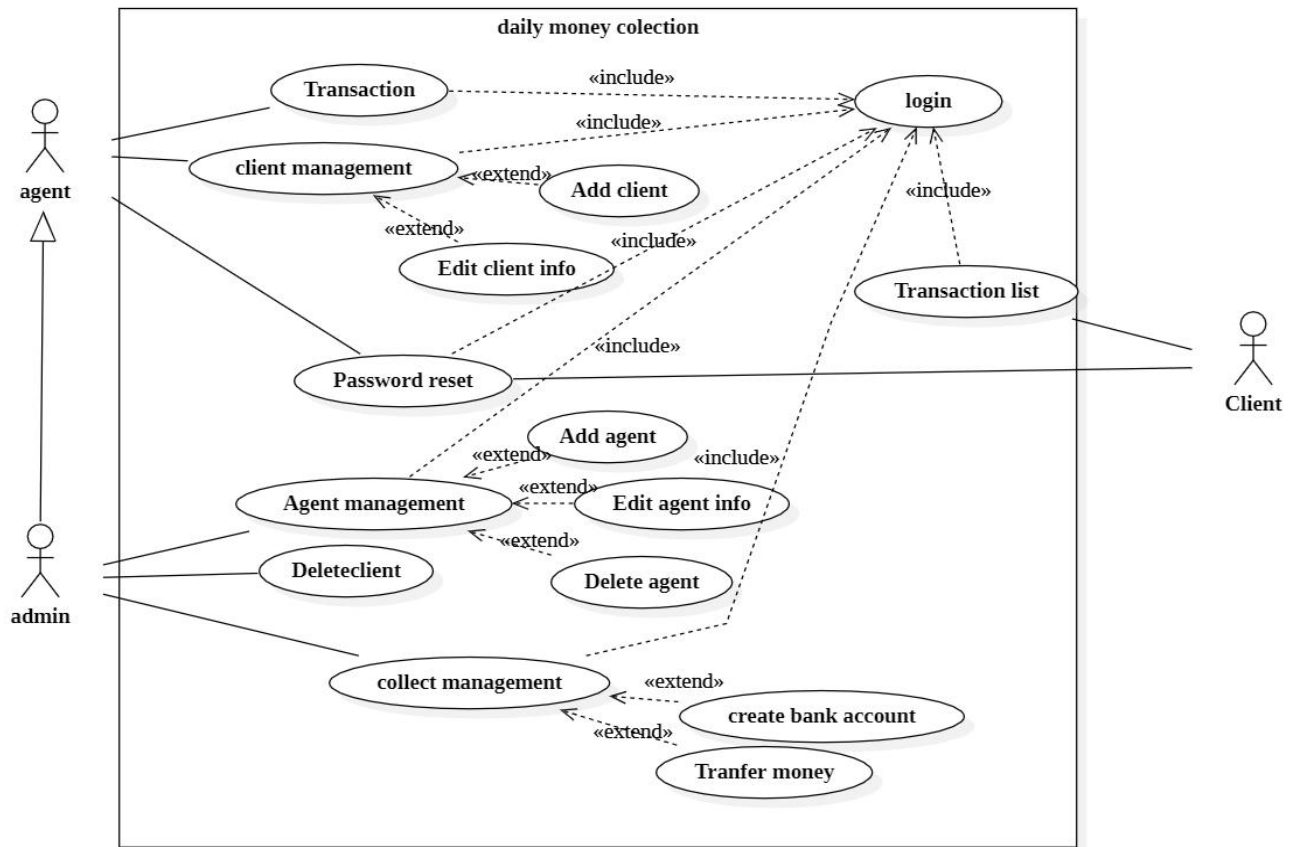


Figure 8: Use case diagram

3.4.2 Sequence diagram

A sequence diagram is an interaction diagram that details how operations are carried out, what messages are sent and when. Sequence diagram are organized according to time. The time progresses as you go down the page. The object involved in the operation are listed from left to right according to when they take part in the message sequence. The purpose of a sequence diagram is to capture the behavior of a system, describe the message flow in the system, structural organization of objects and to describe interaction among objects.

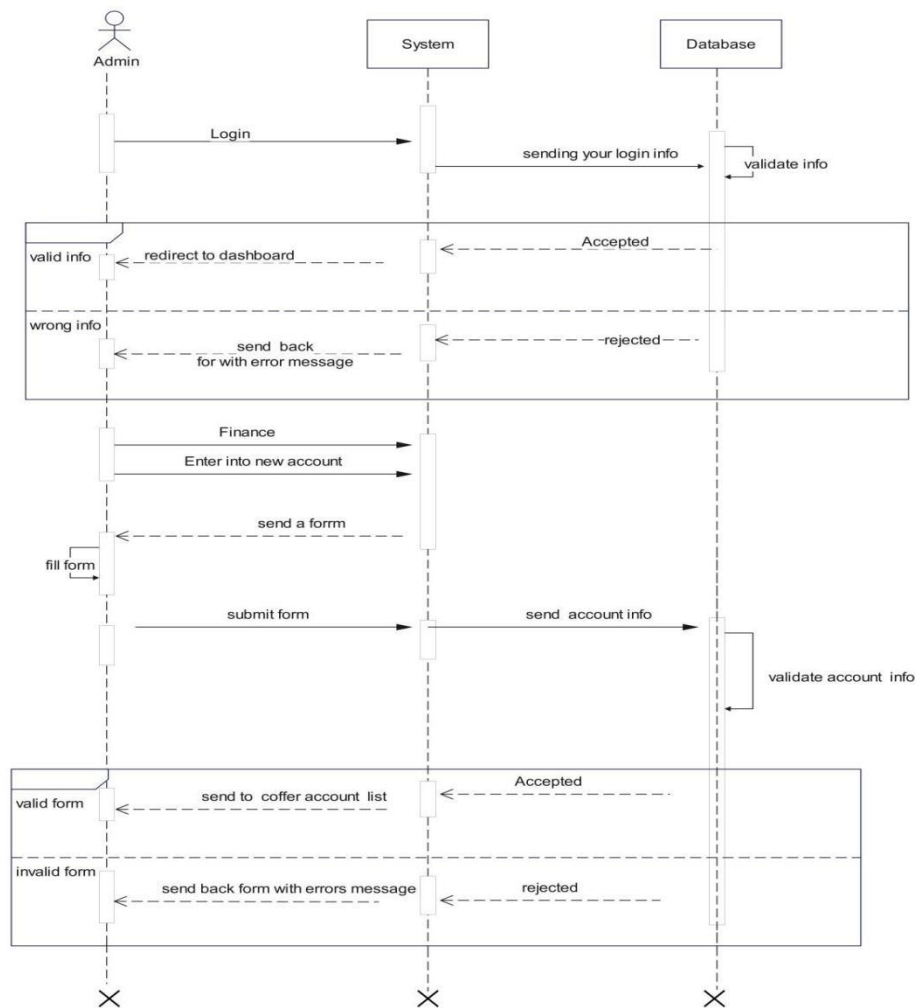


FIGURE 9:CREATE BANK ACCOUNT SEQUENCE DIAGRAM

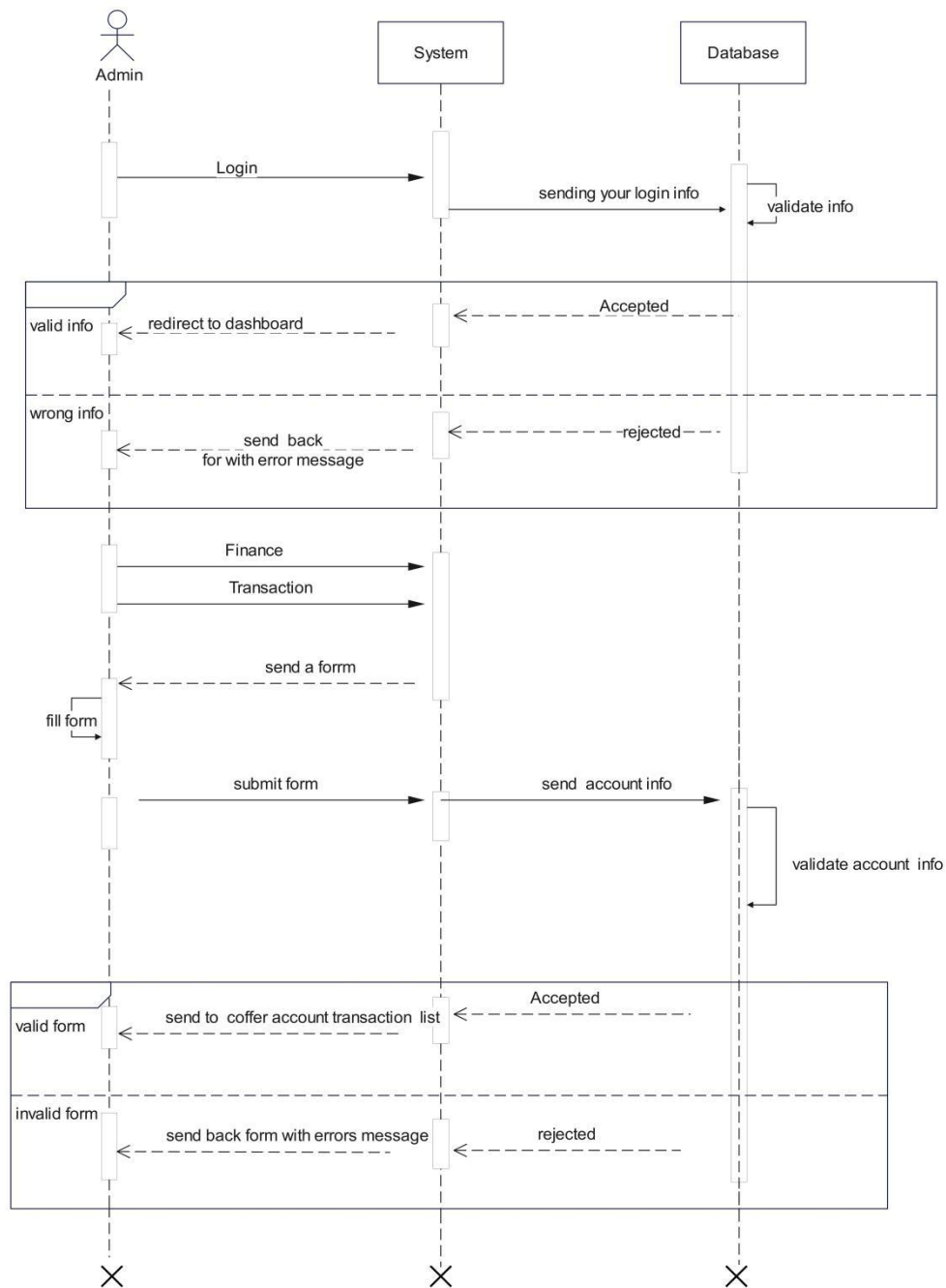


FIGURE 10: TRANSACTION SEQUENCE DIAGRAM

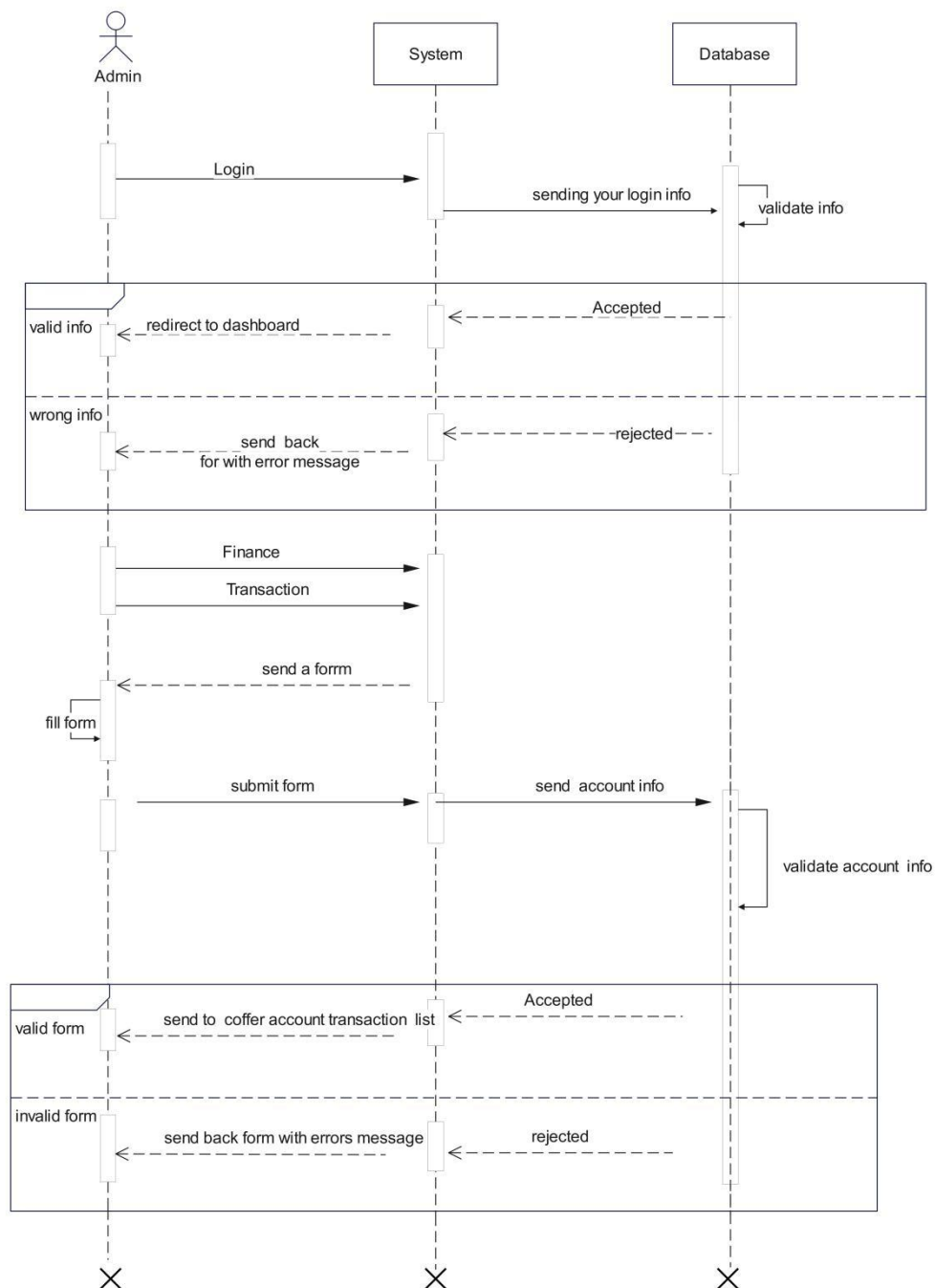


FIGURE 11: COFFER ACCOUNT TRANSACTION SEQUENCE DIAGRAM

3.4.3 STATE CHART DIAGRAM

A state chart diagram is a visual representation of the states of an object and the transitions between those states. It illustrates how an object behaves in response to various events over time. Key components include states, transitions, events, and actions. This type of diagram is commonly used in software engineering to model dynamic behavior. It aids in understanding and documenting system functionalities.

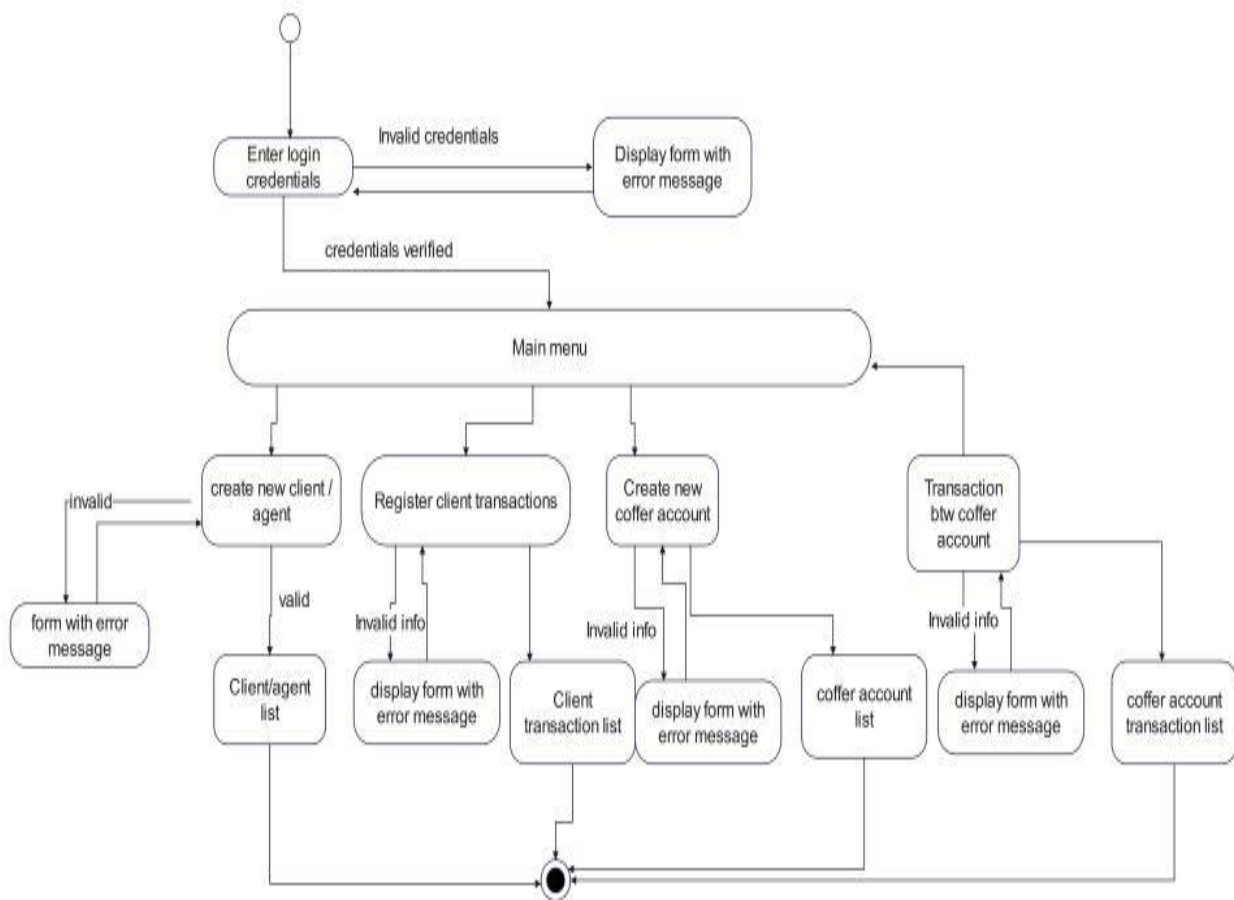


FIGURE 13: SYSTEM STATE CHART DIAGRAM

3.4.3 DEPLOYMENT DIAGRAM

A deployment diagram is a type of diagram used in **Unified Modeling Language (UML)** to visualize the physical deployment of artifacts (like software components, files, or executable programs) on hardware devices or nodes. It shows how various software components (e.g., databases, web servers, applications) interact and are physically deployed on machines, network devices, or servers in a system.

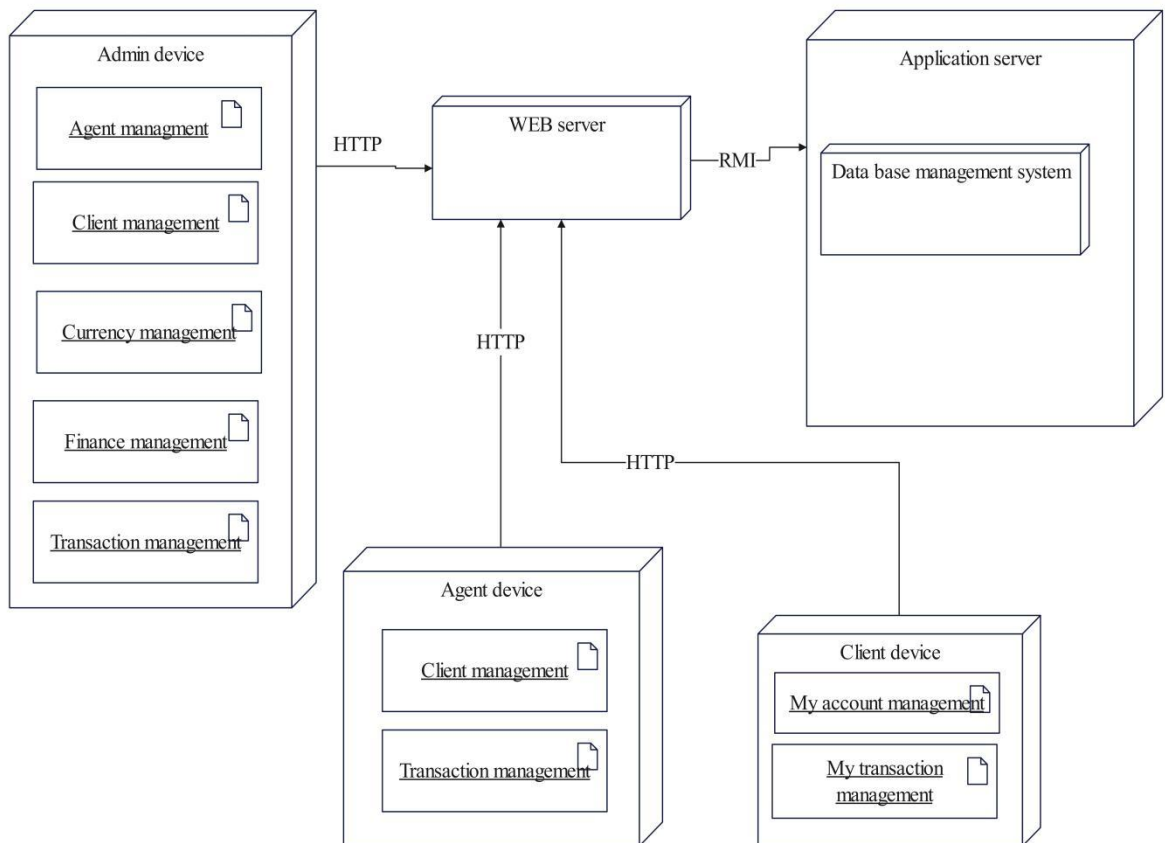


Figure 14:SYSTEM DEPLOYMENT DIAGRAM

3.4.5 CLASS DIAGRAM

A class diagram is used to visualize, describe, document various different aspects of the system, and also construct executable software code. It shows the attributes, classes, functions, and relationships to give an overview of the software system. It constitutes class names, attributes, and functions in a separate compartment that helps in software development

The classes involved in this system are;

USER,ADMIN,AGENT,CLIENT,USER_TYPE,TRANSACTION,Coffer_Account, CURRENCY,Coffer_transaction

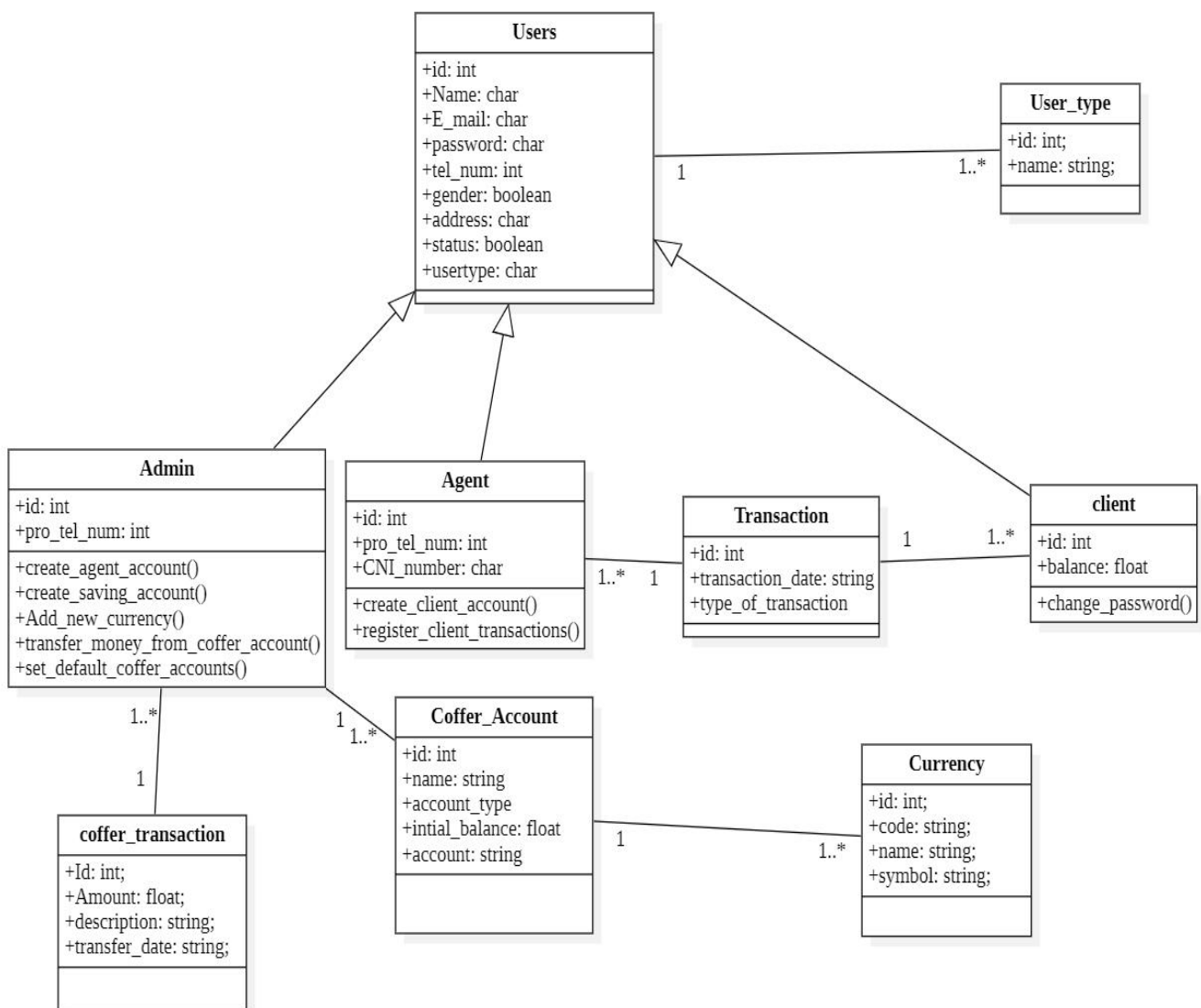
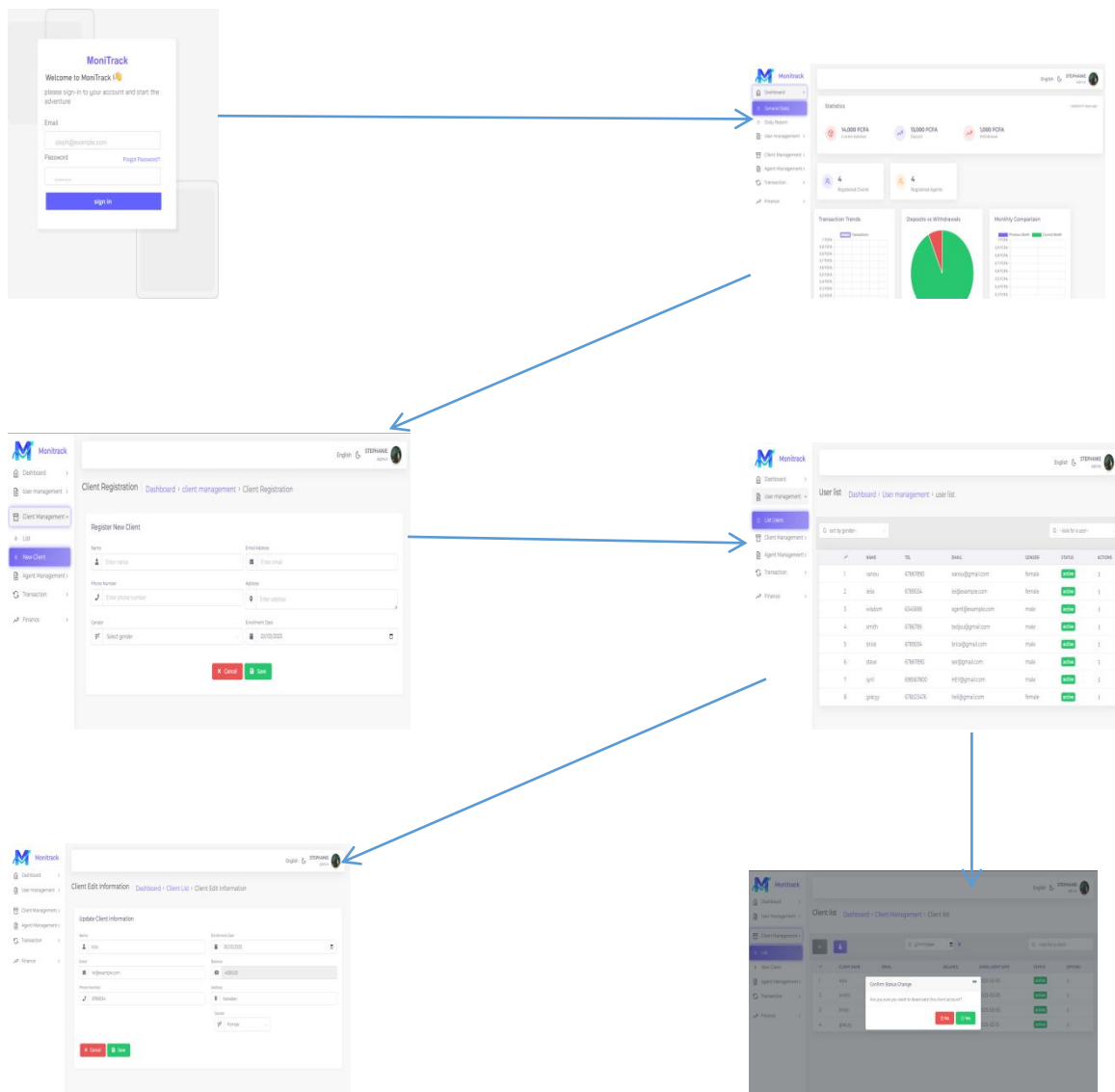


Figure 15:SYSTEM CLASS DIAGRAM

3.4.6 UX/UI DESIGN

The UX/UI design plays a crucial role in enhancing user experience and interaction with software applications. A well-crafted user interface (UI) allows users to navigate through features and functionalities intuitively, making tasks more efficient and enjoyable. Monitrack's UX/UI design incorporates user-friendly elements, clear navigation, and aesthetically pleasing visuals to create an engaging experience. The design prioritizes accessibility and usability, ensuring that users can easily understand and utilize the application's features without extensive training. The design process involved careful consideration of user feedback and best practices in interface design, resulting in a cohesive and functional application.

In the following section, we will present the UX/UI designs created using Figma.



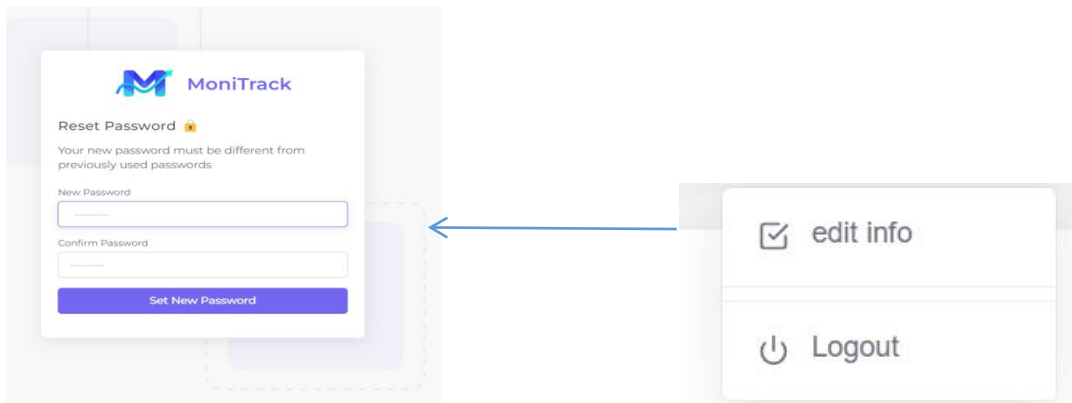


FIGURE 16: UX/UI DESIGN OF MONITRACK

S3.4.7 APP LOGO

The logo of an application serves as a vital element of its brand identity, encapsulating its purpose and values in a single visual mark. For the Monitrack application, we aimed to create a logo that reflects its focus on daily money collection and financial management in a clear and engaging .Using Canva,we have been able to design this logo

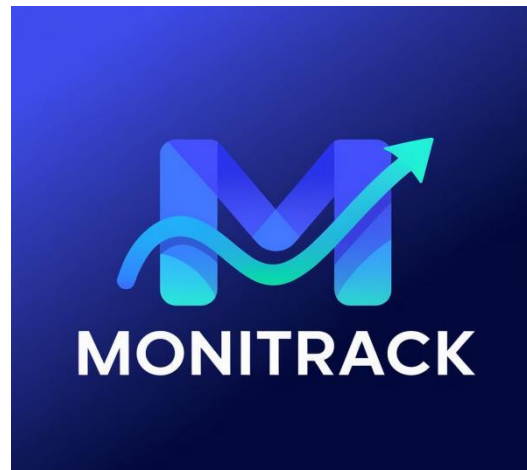


FIGURE 17: APP LOGO

CHAPTER FOUR:IMPLEMENTATION DETAILS AND RESULTS OBTAINED

Following the analysis phase, We moved on to the implementation of the application, which encompasses development aspects. This phase involved defining the software specifications necessary for building the application. I carefully selected the technologies and tools that would facilitate the development process, ensuring compatibility and efficiency. Additionally, I outlined the system architecture and identified any external dependencies. This comprehensive approach allowed for a smoother transition from design to implementation, setting the stage for the successful deployment of the application.

4.1 IMPLEMENTATION DETAILS

4.1.1. TOOLS AND TECHNOLOGIES USED

TYPES OF TOOLS	TOOLS	DESCRIPTION
Database	 XAMPP SERVER	For this project, We decided to use XAMPP for the testing which is a model of web service stacks named as an acronym of its original components: Cross-platform application Apache HTTP Server used as a server for hosting the web application locally during development. It is used to run PHP scripts on local machine. MySQL used to execute Structured Query Language(SQL) on web pages.

Web programming	 Visual studio code	This tool offers an integrated editor for PHP, HTML, Bootstrap CSS, JavaScript, Laravel, and MySQL, featuring real-time code analysis, error detection, and automated refactoring capabilities. Web Browser For this task, we chose to utilize Google Chrome.
Mobile programming	 Android Studio	Android Studio is the official Integrated Development Environment (IDE) for Android app development. It is based on IntelliJ IDEA and was developed by Google specifically for building Android applications. Android Studio provides all the tools needed for building, testing, and debugging Android apps, and it integrates with other Android development tools and services to streamline the development process.
Design		Canva is a user-friendly graphic design platform that allows individuals and businesses to create a wide range of visual content, including social media graphics, presentations, posters, and marketing materials. Launched in 2013, Canva provides an intuitive drag-and-drop interface, making

		design accessible to users without extensive graphic design experience.
design		Figma is a cloud-based design tool that enables collaborative interface design and prototyping for web and mobile applications. Launched in 2016, Figma is particularly popular among UI/UX designers due to its real-time collaboration features and intuitive interface.
Design		StarUML is a powerful UML (Unified Modeling Language) modeling tool designed for software development and systems design. It provides a comprehensive environment for creating various UML diagrams, including class diagrams, sequence diagrams, use case diagrams, and more. StarUML is particularly useful for software architects, developers, and analysts.

TABLE 16:TOOLS USED

LANGUAGE	DESCRIPTION
----------	-------------

 Hypertext Pre-processor	Hypertext Pre-processor is a language specially designed for the development of web applications. It is easy and simple to learn, extremely flexible, easy integration and compatibility, efficient performance, cost efficient, easily connects to database
 HTML BLADE	Blade is a templating engine provided by the Laravel PHP framework. Blade allows developers to write HTML with embedded PHP code, making it easier to create dynamic web applications. It offers features like template inheritance, sections, and control structures (such as loops and conditionals), which simplify the process of generating HTML based on dynamic data.
	CSS (Cascading Style Sheets) is a stylesheet language used to describe the presentation and formatting of a document written in HTML or XML. It enables web developers to control the layout, colors, fonts, and overall visual appearance of a website, separating content from design.
 JAVA SCRIPT	JavaScript is a high-level, dynamic programming language primarily used for adding interactivity and functionality to websites. It enables developers to create responsive user interfaces, manipulate the Document Object Model (DOM), and handle events in real time.
	Java is a high-level, object-oriented programming language designed for portability, security, and performance. Developed by Sun Microsystems (now owned by Oracle), Java is widely used for building platform-independent applications, thanks to its "write once, run anywhere" capability facilitated by the Java Virtual Machine (JVM).

	MySQL is an open-source relational database management system (RDBMS) that uses Structured Query Language (SQL) for managing and manipulating data. It is widely used for web applications and is a key component of the LAMP stack (Linux, Apache, MySQL, PHP/Python).
---	---

TABLE 17:LANGUAGES USED

FRAMEWORK	DESCRIPTION
	Laravel is a free, open-source PHP web application framework that follows the Model-View-Controller (MVC) architectural pattern. It provides a robust set of tools and features for building web applications, ranging from simple websites to complex enterprise-level applications. Laravel emphasizes developer productivity, code simplicity, and elegant syntax. It includes features like routing, caching, database abstraction, authentication, and more, making it easier for developers to build and maintain PHP applications.

 BOOTSTRAP 5	Bootstrap is a widely used front-end framework designed for developing responsive and mobile-first websites. It provides a collection of pre-styled components, such as buttons, forms, and navigation bars, along with a grid system for layout management. Bootstrap enables developers to create visually appealing web applications quickly by using its ready-made classes and components. Its responsive design ensures that applications look great on various devices and screen sizes. With a strong community and extensive documentation, Bootstrap simplifies the process of building modern web interfaces.
--	---

TABLE 18:FRAMEWORK USED

4.1.2 DEPLOYMENT

Deployment is the process of transferring an application from a development environment to a production environment, making it accessible to users. It involves configuring the application, setting up servers, and ensuring that all dependencies are in place. Effective deployment ensures that the application runs smoothly and is maintained efficiently.

Here are a few free servers where you can deploy your application:

1. **Heroku:** Offers a free tier for deploying web applications, supporting several programming languages.
2. **GitHub Pages:** Ideal for static websites, allowing you to host directly from a GitHub repository.
3. **Vercel:** Provides free hosting for front-end frameworks and static sites, with easy integration with Git repositories.
4. **Glitch:** A platform for building and sharing web apps, offering collaborative editing and free hosting.

In our case we used control panel which is requires a paid license .

- ✓ **Control Panel (CPanel):** it is a widely used web hosting control panel that provides a graphical interface and automation tools to simplify the management of web hosting accounts. It allows users to manage their websites, databases, email accounts, and files

with ease. cPanel is particularly beneficial for deploying applications, as it includes features such as file management, database management, and quick access to various server settings..

To deploy using C panel you need the following:

- ◆ **Hosting Account with cPanel Access:** You must have a web hosting account that provides cPanel as part of its services. This account will give you access to manage your website files, databases, and other settings.
- ◆ **Domain Name:** A registered domain name that points to your hosting account. This domain will be used to access your application once it is deployed.
- ◆ **Application Files:** The files of your web application, including source code, configuration files, and any assets (images, stylesheets, etc.). For a Laravel application, ensure you have the complete structure, particularly the public directory.
- ◆ **Database Credentials:** If your application uses a database, you will need the database credentials (hostname, username, password, and database name) to set up the database in cPanel.
- ◆ **FTP/SFTP Client:** A file transfer client like WinSCP or FileZilla to upload your application files to the server. Alternatively, you can use the File Manager in cPanel.
- ◆ **SSL Certificate (Optional):** If you plan to run your application over HTTPS, you will need an SSL certificate. Many hosting providers offer free SSL certificates via Let's Encrypt, which can be managed through cPanel.
- ◆ **Knowledge of cPanel Features:** Familiarity with cPanel's various features, such as File Manager, MySQL Databases, phpMyAdmin, and the Terminal (if available), to effectively manage your deployment.

Deploying monitrack

1. Log in to cPanel:

- Access your cPanel account by entering your domain followed by /cpanel (e.g., www.yourdomain.com/cpanel).
- Enter your username and password to log in.

2. Navigate to File Manager:

Once logged in, locate the **File Manager** icon under the **Files** section. Click on it to open.

3. **Select the Directory:**

- ❖ In the File Manager, navigate to the directory where you want to deploy your application. This is usually in the `public_html` folder for web applications.
- ❖ If you're deploying a subdomain or a specific folder, navigate to that specific directory.

4. **Upload Your Files:**

- ❖ Click on the **Upload** button in the File Manager toolbar.
- ❖ You can drag and drop files or click to select files from your local machine.
- ❖ Ensure that you upload all necessary files, including HTML, CSS, JavaScript, and any other assets required by your application.

5. **Extract Files (if needed):**

- ❖ If you uploaded a zipped file containing your application, right-click on it in File Manager and select **Extract** to unpack the files.

6. **Set File Permissions:**

- ❖ Right-click on your files or directories and select **Change Permissions** to set the appropriate permissions. Ensure that the necessary files are readable and executable.

7. **Create a Database (if applicable):**

- ❖ If your application requires a database, go back to the cPanel home and locate the **MySQL Databases** section.
- ❖ Create a new database and user, then assign the user to the database with the appropriate privileges.

8. **Configure Application Settings:**

- ❖ If your application requires database connection settings, edit the configuration file (often named `config.php` or similar) to include the database name, username, and password.

9. **Testing:**

- ❖ After all files are uploaded and configured, test your application by accessing it through your web browser using your domain name.

10. **Troubleshooting:**

- ❖ If you encounter issues, check error logs in cPanel under the **Metrics** section for more information on what might be going wrong.

Link to my application:<https://monitrack.koralys.cm/first>

Start back the testing phase on the online application.

4.1.3 MOBILE VIEW

A **WebView** is a component that allows you to display web content (like a webpage or web app) directly inside a native mobile application. It essentially acts as a **browser** within your app, but instead of opening a separate browser window, it displays the content as part of the app itself.

WebView is commonly used in **Android** and **iOS** development to integrate web pages, external websites, or dynamic content (such as online forms or media) into a mobile app without leaving the app's interface.

Implementing Monitrack web view:

For the implementation of monitrack Web view we used android studio ,the present web view is available only for android devices



This phase consists of the implementation part of the web application. It's made up of images of web application .

➤ Advantages of using a web view

1. Seamless Integration

Web views are integrated directly within the app, providing a smoother experience without redirecting users to an external browser. This keeps users engaged with the app's core functionalities.

2. Enhanced User Experience

Apps can customize the web view's appearance and behavior, offering a tailored experience that aligns with the app's branding, unlike the generic interface of web browsers.

3. Improved Performance

Web views can be optimized for specific content, potentially leading to faster loading times and better performance compared to a standard browser that loads all features and settings.

4. Offline Access

Web views can cache content, enabling users to access previously viewed information even when they are offline, unlike web browsers which generally require an internet connection.

5. Simplified Navigation

In-app navigation can be more intuitive, allowing users to easily return to previous screens or sections without using the back button in a browser.

6. Reduced Distractions

Using a web view minimizes distractions from browser elements like tabs, bookmarks, and other navigation features, allowing users to focus solely on the content.

7. Consistent Security Model

Apps can implement specific security measures and permissions for web views, providing a more controlled environment for sensitive transactions compared to a browser.

8. Access to App Features

Web views can interact with native app features (like GPS, camera, etc.), allowing for functionalities that are not available in a standard web browser.

9. Faster Updates

Content within web views can be updated without requiring users to update the app itself, ensuring that users always have the latest version of the content.

4.2 RESULT OBTAINED

In this section the result of the various implementation.

4.2.1 MAIN INTERFACES

➤ LOGIN

This login page is for all the users in the system, which are the Admin, Agent, Client all use this page to get into the system.

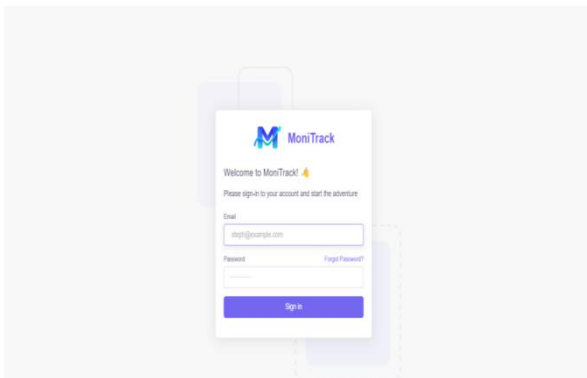


FIGURE 18: LOGIN PAGE

DASHBOARD

This admin dashboard which have all the available functionalities of the application.

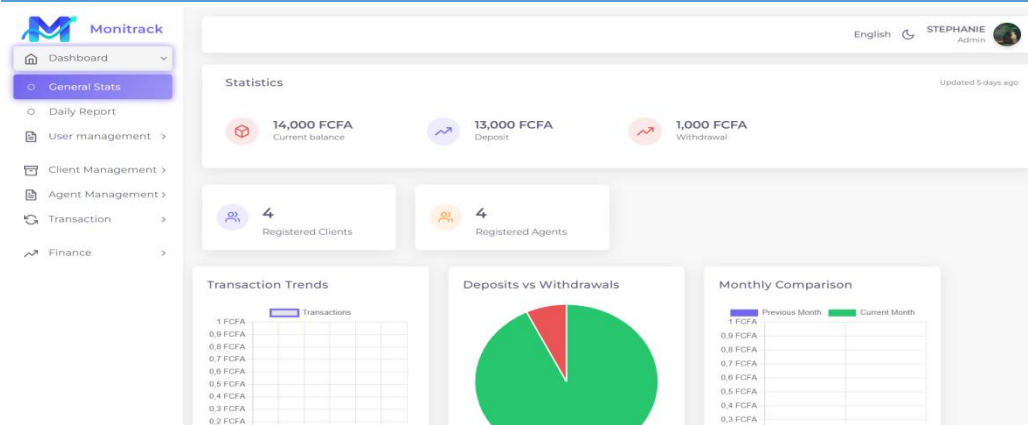


FIGURE 19:DASHBOARD

4.3 ACCEPTANCE TEST CASES

Test case for “ Add for creating a new agent”

IDENTIFICATION: Add Agent

Admin: STEPHANIE

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
Name:vanou Email:vanou@gmail.com Address:ndogpassi Tel:67867890 Gender: female	Admin enter the login credentials(email,password) Upon successful login goes agent management and choose new then enter these credentials on the form and click on save	Send the admin to the Agent list with the new agent on the list	Send the admin to the Agent list with the new agent on the list
RESULT			PASS

TABLE 19:TEST CASE FOR CREATE AGENT ACCOUNT.

Test case for “ Edit agent information”

IDENTIFICATION: Edit Agent(UC-2)

Admin: STEPHANIE

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
Email:vanelle@gmail.c	Admin enter the login	New	New

om	credentials(email,password) Upon successful login goes agent management and choose list then click on option for the agent he wants to edit information and choose edit . A prefilled form is given to be admin and he can update what he wants.	email :vanelle@gmail.com	email :vanelle@gmail.com
RESULT			PASS

TABLE 20: TEST CASE FOR EDITING AGENT INFORMATION.

Test case for “ Delete agent”

IDENTIFICATION: Delete Agent(UC-3)

Admin: STEPHANIE

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
/	Admin enter the login credentials(email,password) Upon successful login goes agent management and choose list then click on option for the agent he wants to edit information and choose delete . A popup message is show for the admin to confirm the delete . Admin click on yes	The agent vanou is not more on the agent list The agent status have changed to inactive in the uses list.	The agent vanou is not more on the agent list The agent status have changed to inactive in the uses list.
RESULT			PASS

TABLE 21: TEST CASE FOR DELETE AGENT ACCOUNT.

Test case for “ Creating a new coffer account”

IDENTIFICATION: BANK(UC-4)

Admin: STEPHANIE

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
Name: MONITRACK Type of account: coffer Initial balance: 2000 Currency: FCFA	Admin enter the login credentials(email,password) Upon successful login goes to FINANCE and choose then click on Account finally take the option new. The admin enters the credentials on the form . Admin click on save	The admin is sent to the coffer account list with the newly created account on the list.	The admin is sent to the coffer account list with the newly created account on the list.
RESULT			PASS

TABLE 22: TEST CASE FOR CREATING A NEW COFFER ACCOUNT.

Test case for “Transferring money between coffer accounts”

IDENTIFICATION: TRANSFER MONEY(UC-5) Admin: STEPHANIE

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
Source Account: MONITRACK Destination account: coffer Amount: 2000 description: test	Admin enter the login credentials(email,password) Upon successful login goes to FINANCE and choose then click on Transfer finally take the option new. The admin enters the credentials on the form . Admin click on save	The admin is sent to the coffer transaction list with the information about the transfer.	The admin is sent to the coffer transaction list with the information about the transfer.
RESULT			PASS

TABLE 23: TEST CASE FOR TRANSFERRING MONEY BETWEEN COFFER ACCOUNTS.

Test case for “Delete client”

IDENTIFICATION: Delete client(UC-6)

Admin: STEPHANIE

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
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/	Admin enter the login credentials(email,password) Upon successful login goes agent management and choose list then click on option for the agent he wants to edit information and choose delete . A popup message is show for the admin to confirm the delete . Admin click on yes	The client lei is not more on the agent list The client status have changed to inactive in the uses list.	The lei is not more on the client list The client status have changed to inactive in the uses list.
RESULT			PASS

TABLE 24: TEST CASE FOR DELETE CLIENT ACCOUNT.

Test case for “Create a client account”

IDENTIFICATION: Add client Agent: wisdom

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
Name:gracyy Email:heli@gmail.com Address:douala Tel:67867893 Gender: female	Agent enter the login credentials(email,password) Upon successful login goes client management and choose new then enter these credentials on the form and click on save	Send the agent to the client list with the new client on the list.	Send the agent to the client list with the new client on the list
RESULT			PASS

TABLE 25: TEST CASE FOR CREATING A CLIENT ACCOUNT.

Test case for “ Register a client transaction”

IDENTIFICATION: TRANSFER MONEY(UC-5)

Agent: wisdom

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
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Client name:smith Type of transaction:deposit Amount:3000 Date:2025-03-05 Amount:2000	Agent enter the login credentials(email:agent@example.com,password:12345678) Upon successful login goes to Transaction and choose New . The admin enters the credentials on the form . Admin click on save	The admin is sent to the transaction list with the information about the transfer.	The admin is sent to the transaction list with the information about the transfer.
RESULT			PASS

TABLE 26: TEST CASE FOR REGISTERING A CLIENT TRANSACTION.

Test case for “ Edit client information”

IDENTIFICATION: Edit client (UC-9)

Admin: STEPHANIE

INPUT	ACTION	EXPECTED OUTPUT	ACTUAL OUTPUT
Email:gracy@gmail	Admin enter the login credentials(email,password) Upon successful login goes agent management and choose list then click on option for the agent he wants to edit information and choose edit . A prefilled form is given to be admin and he can update what he wants.	New email :gracy@gmail.com	Form send back with the message invalid Email format
RESULT			FAIL

TABLE 23: TEST CASE FOR EDITING CLIENT INFORMATION

GENERAL CONCLUSION

The Daily Money Collection Management System was developed to enhance and simplify enterprise management processes. This system is intended to be affordable, straightforward, and user-friendly, enabling rapid calculations and efficient handling of financial data. The system contains multi-features for the effective management of the daily money collect. I learned a lot from developing this project, research done from the internet increased my skills and abilities as a Software developer especially in developing databases and problem design.

Finally, my internship at HONOWA-TECH was very constructive and enriching because one thing led me to learn many other things. I learnt many things in software engineering especially on web development that helped me to come out with the DAILY MONEY COLLECT MANAGEMENT SYSTEM. In conclusion the internship was very good.

PROBLEM FACED DURING THE DESIGN AND IMPLEMENTATION OF MONITRACK

The objectives I had was to build a **DAILY MONEY COLLECT MANAGEMENT SYSTEM** that could manage the real time registration and verification of daily money collection .

I faced challenges in:

a. TECHNICAL:

A lot of time spent during the Designing and Coding phase. I had to learn a specific framework which was not an easy task. It slowed me down for the completion of the system.

b. PERSPECTIVES :

As technology keeps on developing and new requirements come up every day, my work cannot be said to be complete. The work I did meets some requirements and objectives observed at the beginning of the internship. However, it can still be evolved and be improved.

COST AND ESTIMATION

Cost estimation is a critical aspect of project management that involves predicting the financial resources required to complete a project successfully. Accurate cost estimation helps stakeholders make informed decisions, allocate resources effectively, and set realistic budgets and timelines.

Importance of Cost Estimation

1. **Budgeting:** Provides a framework for financial planning and ensures that sufficient funds are allocated for all project phases.
2. **Resource Allocation:** Helps in determining the necessary resources (human, technological, and material) required to achieve project goals.
3. **Risk Management:** Identifies potential financial risks early, allowing teams to implement mitigation strategies.
4. **Performance Measurement:** Establishes a baseline for measuring project performance and determining variances between planned and actual costs.

Cost Estimation Techniques

Various techniques are employed to estimate project costs, including:

- **Analogous Estimating:** Uses historical data from similar projects to predict costs.
- **Parametric Estimating:** Relies on statistical relationships between historical data and other variables (e.g., project size).
- **Bottom-Up Estimating:** Breaks down the project into smaller components and estimates costs for each, then aggregates them.
- **Expert Judgment:** Leverages insights from experienced professionals to provide estimates based on their knowledge.
- **Constructive Cost Model(COCOMO):** It is a widely used model for estimating the cost, effort, and schedule of software development projects. Developed by Barry

Boehm in the early 1980s, it helps project managers and developers assess the resources required for a software project based on various parameters.

- **Function Point Analysis (FPA)** :It is a standardized method for measuring the size and complexity of software applications based on the functionality provided to users. It quantifies software in terms of function points (FP), which represent the functional requirements of the software. FPA is widely used for software cost estimation, project management, and productivity assessment.

For our project we used the function point analysis

Step 1: Identify Functionalities and Assign Weights

Based on function complexity, we assign weights using typical FPA weights

Functionalities	weight
Client management	4
Agent management	4
Transaction management	5
Finance management	5

Step 2: Calculate Function Points

To compute the total function points, we sum the weights assigned to each functionality:

$$\text{Total Function Points} = 4 + 4 + 5 + 5 = 18$$

Step 3: Determine Productivity Rate

Assuming an average productivity rate based on historical data:

- **Function Points per Person-Month: 2.5 FP/person-month**

Step 4: Estimate Effort and Cost

To estimate the total effort required to develop the application, we use the formula:

$$\text{Total Effort (Person-Months)} = \text{Productivity Rate} \times \text{Total Function Points}$$

Substituting the values:

$$\text{Total Effort} = 2.5 \text{ FP/person-month} \times 18 \text{ FP} = 7.2 \text{ person-months}$$

Assuming the average cost per person-month is **300,000 FCFA** (applying Cameroon principles)

Total Cost=Total Effort×Cost per Person-Month

Total Cost=7.2 person-months×300,000 FCFA/person-month=2,160,000 FCFA

Step 5: Consider Cost Drivers and Risks

1. Team Experience:

- An experienced team may reduce effort by **10%**.

2. Complexity of Banking Regulations:

- Assume that regulatory requirements increase effort by **15%**.

1. Experienced Team Adjustment:

Adjusted Effort=Total Effort−(Total Effort×0.10)

Adjusted Effort=7.2−(7.2×0.10)=7.2−0.72=6.48 person-months

1. Regulatory Complexity Adjustment:

Final Adjusted Effort=Adjusted Effort+(Adjusted Effort×0.15)

Final Adjusted Effort=6.48+(6.48×0.15)=6.48+0.972=7.452 person-months

Adjusted Total Cost Calculation

Total Cost=Final Adjusted Effort×Cost per Person-Month

Substituting values: Total Cost=7.452 person-months×300,000 FCFA/person-month=2,235,600 FCFA

Conclusion

In this detailed analysis, after using Function Point Analysis and considering productivity rates, as well as adjusting for cost drivers such as team experience and regulatory complexity, the estimated cost of developing the application is approximately **2,235,600 FCFA**.

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